

LESSON 5.2

PLOTTING ON THE COORDINATE PLANE



LESSON INTRODUCTION

MA.6.GR.1.1 Extend previous understanding of the coordinate plane to plot rational number ordered pairs in all four quadrants and on both axes. Identify the x- or y-axis as the line of reflection when two ordered pairs have an opposite x- and y-coordinate.

LEARNING TARGETS

- I can plot rational ordered pairs.
- I can plot integers using different scales.

BENCHMARK COHERENCE

BUILDING ON

MA.5.GR.2.1 Find the perimeter and area of a rectangle with fractional or decimal side lengths using visual models and formulas.

WORKING TOWARD

MA.7.GR.1.1 Apply formulas to find the areas of trapezoids, parallelograms, and rhombi.

ESSENTIAL QUESTIONS

- How can you graph ordered pairs that have fractions on the coordinate grid?
- How can you graph ordered pairs that are decimals on the coordinate grid?
- How can you determine the best scale for plotting a set of ordered pairs?

LESSON NARRATIVE

In Lesson 1, students plotted points with integer values on one-unit scales. This lesson builds on what students learned previously to include plotting integer coordinates on a set of axes using different scales. During 5.2.2 Refresher, students explore the need for different scales. Students then plot fractions, decimals, and negative numbers accurately on scales that they have created to best fit the given ordered pairs.

COMPONENT SUMMARY

5.2.1 WARM-UP (~5 MIN.)

Interior design career profile

5.2.3 EXPLORATION (10 MIN.)

Plot points using decimals and fractions

5.2.5 GUIDED INSTRUCTION (10-12 MIN.)

Determine the best scale to use and plot rational numbers

5.2.7 WRAP-UP (~5 MIN.)

Plot rational numbers on an accurate scale

5.2.2 REFRESHER (5 MIN.)

A reminder of common, but foundational, key terms.

5.2.4 GUIDED PRACTICE (5 MIN.)

Practice plotting rational numbers

5.2.6 COLLABORATION (5-8 MIN.)

Work with a partner to find best scale for a data set

RESOURCES

GLOSSARY

Key Terms:

- Coordinate plane
- Axes
- Coordinate
- Origin
- Quadrant
- Scale

Additional Terms: N/A

MATERIALS

- (Optional) Paper
- Graph paper

ADDITIONAL PREPARATION

- 5.2.1 Warm-Up contains a video to accompany the question prompts.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may assume they can just use a scale of one unit when creating the graphs, resulting in not being able to fit all of the coordinates in the plane.
- Students may incorrectly plot negative points in the wrong quadrant.
- Students may struggle with estimating the location of rational ordered pairs.

THINGS TO CONSIDER

- Determine if your students need the support of all or part of the Refresher.
- Help students create appropriate graphs and scales by asking for the largest and smallest x and y values that need to fit in the allotted space.
- Remind students that the x-axis and y-axis do not need to be drawn in the center of the grid.
- Students are still grasping negative numbers. Consider providing vertical and horizontal number lines for students to reference.
- If time is a constraint, assign students to work through 5.2.6 Collaboration for homework.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

During 5.2.4 Guided Practice, engage students in a Think-Pair-Share routine to encourage students to discuss and build upon their reasoning for choosing an appropriate graph and scale to represent the data. This will encourage students to explain why their selection is reasonable and formalize their responses with a partner before engaging in the written explanation.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.6.1 Reasonableness

5.2.4 Guided Practice provides an opportunity for students to assess the reasonableness of three potential coordinate graphs to represent a data set. Students must apply what they know about appropriate scales to determine the best coordinate plane to plot the data points and explain why the other options are not as accurate. Through this activity, students are encouraged to verify their solutions and evaluate the potential answers in the given context of the data. Encourage students to consider whether their chosen coordinate graph makes sense and justify why.

DIFFERENTIATE INSTRUCTION

Consider providing Desmos activity “The (Awesome) Coordinate Plane Activity” as support for remediating student conceptual understanding of plotting points on the coordinate plane. This interactive digital activity provides students with additional practice in plotting points on the coordinate plane, as students change the x- and y-value of points to hit the bullseye on a target. As the activity progresses, students practice hitting the target with different scales and then estimate and predict the coordinates of the bullseye given two other data points without scale.



5.2.1 WARM-UP: INTERIOR DESIGNER

Component Overview: In this Warm-Up, students will learn about a career in interior design. After watching a video showing Soledad Alzaga discussing her career, students will consider what career they are interested in and what skills they need for that career.

QUESTIONING

Do you know someone who is an interior designer or works in graphic design? Consider bringing in a visitor for your class to ask questions and learn from. Additionally, consider having students return to this lesson later in the year for project-based learning around this Career Profile.



5.2.1 Warm-Up: Interior Designer

After getting a degree in graphic design, Soledad Alzaga built her career and company as an interior designer. Interior designers make interior spaces functional, safe, and beautiful by determining space requirements and selecting decorative items such as colors, lighting, and materials. Beyond being creative and having good taste, Soledad says to own a successful interior design business, it is necessary to have the ability to work and communicate with others, be organized, have business knowledge, and to have math and measurement skills.

1. What is something that interests you that you would like to spend your career doing?

Answers will vary. I am interested in becoming a Youtuber/Influencer.

2. What skills do you think are needed to be successful in that career?

Answers will vary. I need to have marketing skills and skills for making videos.



5.2.2 Refresher: Key Terms with the Coordinate Plane

Several key terms related to the coordinate plane were used throughout Lesson 1. Below, you can find definitions for each of these key terms.



The **coordinate plane** is a plane determined by two perpendicular number lines, called axes. The axes intersect at the origin. Each point in the coordinate plane is represented by a pair of coordinates that represent the direction and distance from each axis. The origin is represented by the coordinate pair, $(0, 0)$.



The **axes** of a graph are the horizontal and vertical number lines used in a coordinate plane system.



A **coordinate** is a number used to locate a point on a number line. A coordinate is one of the numbers in an ordered pair that locates a point on a coordinate plane.



The **origin** is the point of intersection of the x - and y -axes in a rectangular coordinate system, where the x -coordinate and y -coordinate are both 0.



A **quadrant** is any of the four regions separated by the axes in a coordinate plane.

5.2.2 REFRESHER: KEY TERMS WITH THE COORDINATE PLANE

Component Overview: Students review key vocabulary from Lesson 1.

DIFFERENTIATE INSTRUCTION

Consider vocabulary strategies that align with your classroom and your students' learning needs.

- Frayer Model
- Word wall
- Anchor chart

TEACHER MOVES

Prompt students to engage in increasingly precise mathematical discussions using the vocabulary provided. The following might be common, informal examples of words to replace:

- Replace "center" with "origin."
- Replace "part" with "quadrant."
- Replace "lines" with "axes."

5.2.3 EXPLORATION: PLOTTING ORDERED PAIRS

Component Overview: Students will work with a partner to discover how to plot fractional and decimal points on a graph. Students will work with a partner to determine different scales that can be used for a graph. Students will explore how to make scales using negative numbers.

DIFFERENTIATE INSTRUCTION

Consider providing a separate copy of the dartboard since it will be used across multiple questions. For students who struggle with differentiating the colors, have them label each ring of the dartboard with the name of the color.

ENRICHMENT

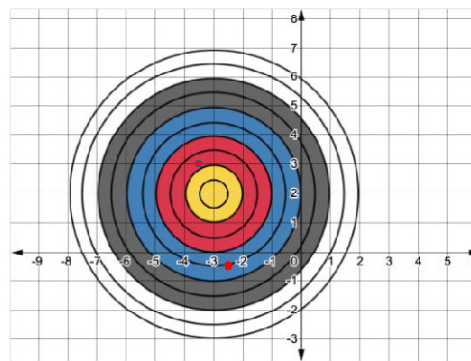
Students may struggle to approximate the coordinates or determine which scale(s) are the most accurate for the coordinate plane. Ask questions to provide further scaffolding like:

- What type of number does the coordinate need to be if the red area does not fall on a whole number?
- What number is halfway between 1 and 2? 2 and 3?
- In question 3, why would a fractional scale be the most accurate for the dartboard activity?



5.2.3 Exploration: Plotting Ordered Pairs

Two players engage in a game of darts.



1. The first player throws a dart that lands at a point with an x -value halfway between -2 and -3 in the blue area. Without looking at your partner's paper, draw a point on the dart board where the dart may have landed.

- a. What are the approximate coordinates of the dart?

Answers will vary. $(-2.5, -0.5)$

- b. What are the approximate coordinates of your partner's dart?

Answers will vary.

2. The second player throws a dart that lands in the red area, but not on one of the black circles. Without looking at your partner's paper, draw a point on the dart board where the dart may have landed.

- a. What are the approximate coordinates of the dart?

Answers will vary. Possible solutions: $(-3.5, 3)$ or $(-2.5, 3)$ or $(-1.25, 2.25)$

- b. What are the approximate coordinates of your partner's dart?

Answers will vary.

3. The scale used on the grid is 1 unit. Discuss with your partner how the scale used on the grid could be changed to make coordinates of the darts more accurate. Give each scale in the table a rating by circling the face that best describes the scale's accuracy, where 😞 means the location of the coordinates would not be very accurate, and 😊 means the location of the coordinates would be very accurate.

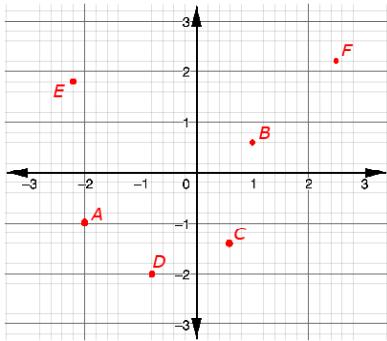
Answers will vary.

Scale	Accuracy Rating		
2	😞	😐	😊
1	😐	😐	😊
current scale	😐	😐	😊
$\frac{1}{2}$	😐	😐	😊
$\frac{1}{4}$	😐	😐	😊



5.2.4 Guided Practice: Plotting Rational Ordered Pairs

1. On the coordinate plane, plot and label the points given.



Point A (-2, -1)	Point B (1, 0.6)	Point C $(\frac{3}{5}, -1\frac{2}{5})$
Point D (-0.8, -2)	Point E $(-\frac{11}{5}, 1\frac{8}{10})$	Point F (2.5, 2.2)

5.2.4 GUIDED PRACTICE: PLOTTING RATIONAL ORDERED PAIRS

Component Overview: Students plot and label rational numbers on a coordinate plane with a determined scale.

DIFFERENTIATE INSTRUCTION

If students are struggling with how they should graph negative and positive coordinates, encourage students to draw arrows above each number to help plot correctly as well as label x and y on each coordinate.

It may also be helpful to provide two number lines and ask the student to plot the x-coordinate on one of the number lines and the y-coordinate on the other number line. Then, have the student make these into a set of axes and find where these two points would meet on the coordinate grid.

ENRICHMENT

Students may struggle to graph improper fractions and decimals because the scales on the x- and y-axes are one. If you notice students struggling, ask guiding questions like:

- If a point in an ordered pair is an improper fraction, what do you need to do first before you can graph it?
- How can you accurately graph a number that is a decimal? Which two whole numbers is this decimal between? Which whole number is it closer to? How can you show that?
- How can you accurately graph a number that is a fraction? What two whole numbers is this fraction between? How do you know?

5.2.5 GUIDED INSTRUCTION: CHOOSING THE RIGHT SCALE

Component Overview: In this Guided Instruction, students work with a partner to determine appropriate scales for graphs based on the data presented. Students choose a graph to best represent a data set and explain why their choice is the best and why the other graphs are not as accurate. Students then determine the appropriate scale to plot given points on a coordinate plane.

TEACHER MOVES

MA.K12.MTR.6.1: Reasonableness

Students will discuss why the given graph is not appropriate for the data with a partner. Prompt students with questions like:

- What is the scale given on the graph?
- How would the graph look when the points are plotted?
- When graphed, would the data accurately show the growth?
- What would be a good scale to use for this set of data?



5.2.5 Guided Instruction: Choosing the Right Scale

When representing data, the scale of the grid is important to effectively and accurately communicate information.

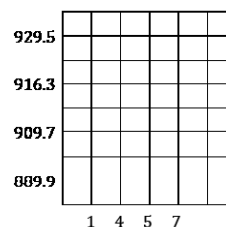


A **scale** is the numeric values, set at fixed intervals, assigned to the axes of a graph.

1. The table shows the total Florida tourism industry employees, in thousands, for years since 2002.

Years Since 2002	Total Tourism Industry Employees (in thousands)
1	889.9
4	909.7
5	916.3
7	929.5

To plot the data, Andrea drew the coordinate grid shown.



Discuss with your partner why Andrea's coordinate grid would not effectively nor accurately communicate the information about Florida tourism industry employees.

Answers will vary. The scale would not accurately communicate information because it is not a consistent scale. For example, some of the units on the x-axis represent 1 year but some of the units represent 3 years.

TEACHER MOVES

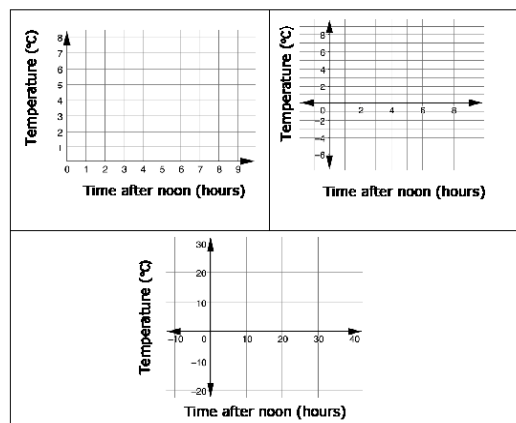
Think-Pair-Share

Leverage the Think-Pair-Share instructional routine to facilitate meaningful discourse using the prompts in questions 2a and 2b below. Prompt students to individually write their responses to parts a and b before sharing with a partner. Give students the opportunity to refine their responses based on what their partner shared before eliciting responses from the whole group. Specifically, listen for students who changed their responses based on what their partner shared and have those students share to continue encouraging growth mindset.

2. The data shown in the table are average autumn temperatures, in degrees Celsius ($^{\circ}\text{C}$), in Carmathen, Wales.

Time After Noon (hours)	Temperature ($^{\circ}\text{C}$)
0	5
1	4
2	3
3	2
4	1
5	-2
6	-3
7	-4
8	-4

Three coordinate grids are shown, each using different scales.



- a. Explain which grid most accurately represents the data.

Answers will vary. The grid in the upper-right corner most accurately represents the data because the x-axis uses a scale that includes all numbers between 0 through 8 while the y-axis includes all the numbers between -4 and 5, which are the ranges of values listed in the table above.

- b. Explain why the other two grids may not be as accurate.

Answers will vary. The grid in the upper left corner is not accurate because it does not include negative values for temperature. The grid at the bottom is not accurate because the scale is too large to accurately plot any of the data and to see any relationships or trends.

When creating an appropriate graph, it is helpful to consider the following.

- Determine which quadrants are needed by identifying the smallest and largest points of each variable.
- Use these values to determine a good scale for the axis that represents the variable.
- Then, determine the best scale for the grid lines by considering the numbers presented in the data.
- Remember that the scale for each axis can be different, but that each axis must have a consistent scale.

3. Four coordinates are given.

$$\left(1, 2\frac{1}{2}\right) \left(3\frac{1}{2}, -4\right) (-5, -2) (0, 3)$$

- a. What is the smallest x-coordinate?
-5

- b. What is the largest x-coordinate?

$$3\frac{1}{2}$$

- c. What is the smallest y-coordinate?

$$-4$$

- d. What is the largest y-coordinate?

$$3$$

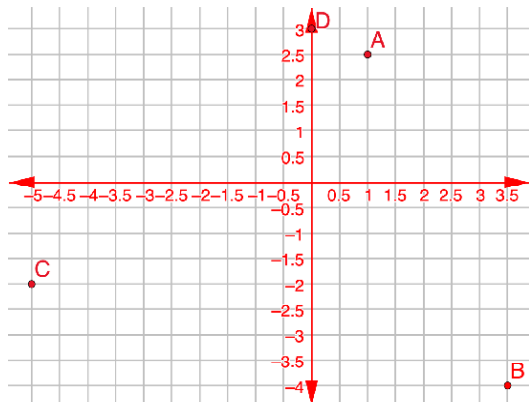
- e. Which quadrants need to be represented on the grid?

All 4 quadrants

- f. By looking at the numbers that make up the ordered pairs, what scale should be used to number each axis?

	x-axis	y-axis
Scale	$\frac{1}{2}$	$\frac{1}{2}$

- g. On the grid, draw and label coordinate axes with an appropriate scale, and plot the points.



5.2.5 GUIDED INSTRUCTION: CHOOSING THE RIGHT SCALE (CONTINUED)

ENRICHMENT

As students consider the coordinate grids shown, circulate and ask questions to aid in their analysis like:

- Look at each graph. Are there any points you cannot plot?
- When you graph the data, does it look different on each graph? Does this matter? Why?
- What do you notice about the scales on each of these coordinate grids? Why might that be important when plotting the data points?

TEACHER MOVES

Think-Pair-Share

Engage students in a Think-Pair-Share routine as they explain questions 3a and 3b. First, have students consider each question on their own. Then students explain their response to a partner and revise or build upon their initial response. This will give students time to reflect on their choices and consider why one graph is more accurate than the rest.

DIFFERENTIATE INSTRUCTION

Students may create a picture guide for the given bullet points to allow an entry point for visual learners. Students may use this to refer to later in the lesson and in future lessons when creating graphs.

5.2.6 COLLABORATION: CHOOSING THE RIGHT SCALE

Component Overview: Students will work with a partner to create a scale on the graph for the given data points and accurately plot rational ordered pairs. Partners will then discuss with another group how they made their scale and analyze any differences between their graphs.

ENRICHMENT

Consider asking students the following:

- What are the lowest and highest x -values?
- What are the lowest and highest y -values?
- What factor of these values could you count by to get from the lowest number to the highest number that allows for numbers to be spread out enough without being too far apart?

DIFFERENTIATE INSTRUCTION

Consider providing students with a separate, larger graph when graphing fractional points in order to allow students to have room to accurately graph each point.

TEACHER MOVES

Consider partnering groups by who graphed differently to make sure students are able to discuss differences in their graphs and how the points would still look the same on the graph even with a different type of scale.



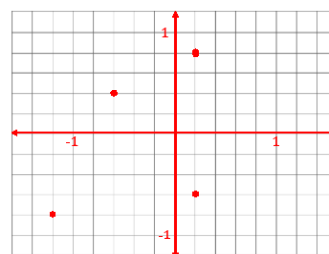
5.2.6 Collaboration: Choosing the Right Scale

1. With your partner, use the given sets of ordered pairs to draw and label coordinate axes with an appropriate scale, and plot the points.

- a. $(50, 50)$, $(0, 0)$, $(-10, -30)$, $(-35, 40)$



- b. $(\frac{1}{5}, \frac{4}{5})$, $(\frac{-3}{5}, \frac{2}{5})$, $(-1\frac{1}{5}, -\frac{4}{5})$, $(\frac{1}{5}, -\frac{3}{5})$



2. Find another group of partners to compare your graphs with. Explain what is the same and different between your graphs and their graphs.

Answers will vary. We used the same scale. We numbered each of the lines by $\frac{1}{5}$. The other group used decimals as a scale. The other group's x -axis was on a different line.

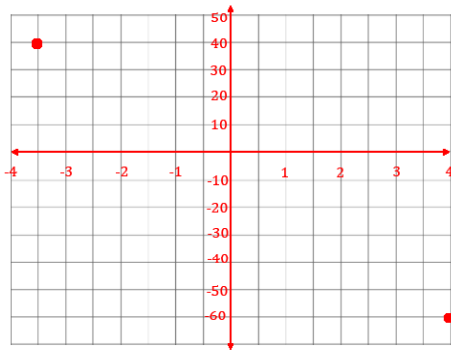
3. If your graphs were different, were both still correct representations of the ordered pairs?

Answers will vary. If students used decimals, the ordered pairs would still be correctly represented.



5.2.7 Wrap-Up: Ticket Out the Door

- When plotting $(-3.5, 40)$ and $(4, -60)$, describe how you would scale the axes.
Answers will vary. First, consider the x-coordinates of both pairs. One coordinate has a .5 therefore scale of the x-axis should be 0.5 or $\frac{1}{2}$. The y-coordinates of both pairs are multiples of 10 therefore the scale for the y-axis should be 10.
- For the points $(-3.5, 40)$ and $(4, -60)$, draw and label coordinate axes with an appropriate scale, and plot the points.



5.2.7 WRAP-UP: TICKET OUT THE DOOR

Component Overview: In this Wrap-Up, students will show their understanding of the lesson. Students will prove they can find the scale and plot rational numbers correctly. Consider using 5.2.7 Wrap-Up as a formative assessment.

DIFFERENTIATE INSTRUCTION

If students struggle with writing small enough on the given graph, consider allowing students to plot the points using a graphing utility.

QUESTIONING

Consider other resources for additional enrichment opportunities for students to practice or test their abilities. The Desmos activity “The (Awesome) Coordinate Plane Activity” provides students opportunities for additional practice plotting points and extending their understanding of coordinate planes.